

**FROM THE ARITHMETIC OF $H(t) = 1 + \sum_n \tau(p_n) t^n$
TO THE DURABLE PERIODICITY OF THE DEFORMED
MINIMUM MODULI
(FIGURES 2–5) — REVISED WITH THE $n \leq 600$ RUN**

CLAUDE

ABSTRACT. We assemble the argument that the oscillation in the deformed minimum-modulus plots for $h_n = \tau(p_n)$ is *durable* as n grows, and characterise it. The chain: (i) Deligne’s bound makes H analytic on the unit disc; (ii) the deformation multiplies the generating function by e^{ct} , so the deformed coefficients are governed by the zero $t_0 = \rho e^{i\theta}$ of H nearest the origin; (iii) singularity analysis gives a single-frequency law $h_n^{(c)} = C_c \rho^{-n} \cos(n\theta + \varphi_c) + (\text{lower order})$; (iv) since the product of the roots of the monic characteristic polynomial is $n! h_n^{(c)}$, the minimum modulus is driven by $|h_n^{(c)}|$; (v) the relative error of the law decays *geometrically*, so the oscillation is the asymptotic law, not a transient — the durability; (vi) the rigid two-gap pattern of the extrema is the return-time structure of the circle rotation by $\alpha = \theta/\pi$, with fast period $1/\alpha \approx 4.11$ and envelope 37 from the convergent $9/37$. This revision incorporates two settled facts. First, the rotation number was pinned to $\theta/\pi = 0.243183594574493\dots$ with continued fraction $[0; 4, 8, 1, 11, 2, \dots]$ — irrational to all computable precision, **not** $9/37$, of which it is the third convergent. Second, the envelope run to $n = 600$ (adaptive precision, $\sim n$ bits) confirms the $\{4, 5\}$ Sturmian word with the 5’s on an exact period-9 schedule throughout, an FFT period of 4.11 at $22\times$ the red-noise threshold, and $0 < \mu_n^{(1)} \leq 51.7$ with dips to 0.066; the predicted second-order defect near 440 has *not* appeared by 600, relocating it beyond the current range. The two analytic load-bearing assumptions (a transfer hypothesis (S) and the dominant complex zero) remain unproved and are flagged as such.

1. OBJECTS AND THE ARITHMETIC INPUT

Let τ be Ramanujan’s function, $\Delta(q) = q \prod_{k \geq 1} (1 - q^k)^{24} = \sum_m \tau(m) q^m$, p_n the n -th prime, and fix

$$h_0 = 1, \quad h_n = \tau(p_n) \quad (n \geq 1), \quad H(t) = \sum_{n \geq 0} h_n t^n, \quad J = t(\log H)'.$$

Equation (D) of [1], $nh_n = \sum_{r=1}^n j_r h_{n-r}$, is the bijection $(h_n) \leftrightarrow (j_r)$, i.e. $tH' = HJ$. *Deformation* by c prepends c to the j -data, multiplying the

generating function by an exponential:

$$H^{(c)} = e^{ct}G, \quad G = \exp \int_0^t J, \quad h_n^{(c)} = [t^n]e^{ct}G = \sum_{k=0}^n \frac{c^k}{k!} g_{n-k}, \quad g_m = [t^m]G.$$

Theorem 2.3 of [1] attaches the monic

$$(1) \quad \chi_n^{(c)}(x) = \sum_{r=0}^n \binom{n}{r} r! h_r^{(c)} (-1)^r x^{n-r}, \quad \mu_n^{(c)} = \min\{|\zeta| : \chi_n^{(c)}(\zeta) = 0\}.$$

Figures 2–5 plot $(n, \mu_n^{(c)})$; the argument is uniform in $c > 0$, the run below being $c = 1$.

Proposition 1.1 (radius of H ; RIGOROUS). *H is analytic on $|t| < 1$.*

Proof. Deligne [7] gives $|\tau(p)| \leq 2p^{11/2}$, and $p_n \sim n \log n$ [8], so $|h_n|^{1/n} \rightarrow 1$ and the radius is ≥ 1 by Cauchy–Hadamard [9]. $\square$

2. REDUCTION TO ONE COMPLEX NUMBER

Because $J = tH'/H$, the singularities of $G = \exp \int J$ in the disc are the zeros of H [9, 10]. Near a simple zero t_0 , $J \sim t_0/(t-t_0)$, $\int J \sim t_0 \log(t-t_0)$, $G \sim (t-t_0)^{t_0}$: a branch point. Hence the g_m (and so the $h_n^{(c)}$) are governed by the zero of H nearest the origin [12].

Observation 2.1 (the dominant zero; EMPIRICAL). Truncating H at degrees 90, 105, 120 and solving, the nearest zero is a stable, simple, isolated conjugate pair $t_0 = \rho e^{i\theta}$, $\rho \approx 0.0435$, $\theta \approx 0.2432\pi$, with the next zero at $\rho_2 \approx 0.078$ [4]; corroborated by $|g_m|^{1/m} \rightarrow 1/\rho \approx 23$. *Not proved*: that this is a genuine simple dominant zero of the full H .

3. THE SINGLE-FREQUENCY COEFFICIENT LAW

Hypothesis 1 (analytic transfer, S). Singularity analysis [11, 12, 13] applies to G at $t_0, \bar{t}_0$: with $\rho_2 > \rho$, $g_m = C\rho^{-m} \cos(m\theta + \varphi) + O(\rho_2^{-m} m^B)$, $C > 0$.

The Flajolet–Odlyzko transfer theorem, valid once G is Δ -analytic with branch points of algebraic–logarithmic type; expected here (t_0 isolated, interior, a genuine branch point) but not verified for this G .

Lemma 3.1 (RIGOROUS GIVEN (S)). $h_n^{(c)} = C_c \rho^{-n} \cos(n\theta + \varphi_c) + O(\rho_2^{-n} n^B)$, with $C_c = C e^{c\rho \cos \theta} > 0$, $\varphi_c = \varphi - c\rho \sin \theta$ [5].

So the deformed coefficients *grow* geometrically and carry a phase $e^{in\theta}$. The discriminating point: a *real* nearest singularity (as for $h \equiv 2$, $G = 1/(1-t^2)$) gives bounded coefficients and a true flat plateau, and a coefficient sequence tending to a constant degenerates $\chi_n^{(c)}$ to a scaled truncated exponential $E_n(-x)$ with $\mu_n^{(c)} \sim \kappa n$ ($\kappa = 0.278465\dots$), an unbounded Szegő ramp [25, 26, 27, 2]. An interior *complex* zero gives neither: it gives oscillation.

4. FROM COEFFICIENTS TO MINIMUM MODULI

Proposition 4.1 (exact criterion; RIGOROUS). *The product of the roots of (1) is $(-1)^n \chi_n^{(c)}(0) = n! h_n^{(c)}$; the product of the moduli is $n! |h_n^{(c)}|$. Hence $\mu_n^{(c)} = 0 \iff h_n^{(c)} = 0$ and $\mu_n^{(c)} \leq (n! |h_n^{(c)}|)^{1/n}$ [9].*

Observation 4.2 (tracking; EMPIRICAL). With the bulk-root product smooth in n , $\mu_n^{(c)}$ tracks $|h_n^{(c)}|$ and dips toward zero at the sign changes of $h_n^{(c)}$ — now seen directly as the dip to $\mu_n^{(1)} = 0.066$ in the $n \leq 600$ run (§7) [4].

5. DURABILITY OF THE OSCILLATION

Theorem 5.1 (durability; CONDITIONAL ON (S), OBS. 2.1). *The single-frequency law is asymptotic: its relative error is $O((\rho/\rho_2)^n n^B) \rightarrow 0$ geometrically ($\rho/\rho_2 \approx 0.56$). Off shrinking windows around the cosine zeros, $|h_n^{(c)}|/(C_c \rho^{-n}) \rightarrow |\cos(n\theta + \varphi_c)|$, so the periodic structure of the driver $h_n^{(c)}$ strengthens with n .*

Remark 5.2 (precision-proofing). The coefficients of (1) span $> 10^{2000}$ by $n = 500$; at fixed 100-bit precision the smallest root is the fragile one, and for the *undeformed* sequence this manufactured a spurious plateau in place of the ramp κn [2]. Two facts insulate the present conclusion. (a) The driver $h_n^{(c)} \in \mathbb{Q}$ is exact (integral j -recurrence), so its sign-change pattern — which sets the dips and the period — is precision-independent. (b) $\mu_n^{(c)} = 0 \iff h_n^{(c)} = 0$ is exact (Prop. 4.1). The magnitude of $\mu_n^{(c)}$ still needs honest root-finding: the $n \leq 600$ run (§7) uses *adaptive* precision ($\sim n$ bits), and the internal consistency of its FFT, ACF and two independent envelope analyses supports robustness of the smallest root for this sequence; a formal certificate is the precision-doubling check, never fixed 100 bits [28, 29].

6. THE ROTATION NUMBER, SETTLED

Pinning θ to high precision (two truncation/precision settings in agreement) gives

$$(2) \quad \alpha := \theta/\pi = 0.243183594574493\dots, \quad \alpha = [0; 4, 8, 1, 11, 2, \dots],$$

irrational to all computable precision and **not** $9/37$ (they differ at the fourth decimal). The convergents are $\frac{1}{4}, \frac{8}{33}, \frac{9}{37}, \frac{107}{440}, \dots$, so $9/37$ is the *third* convergent — the 9 is the numerator p_3 , not a partial quotient. This is what the period-37 envelope was tracking, and it removes the “exactly $9/37$ forever” scenario [4, 6, 8, 24].

7. WHAT THE $n \leq 600$ RUN SHOWS

The deformed $c = 1$ minimum moduli were computed to $n = 600$ at adaptive precision and the lower envelope difference-tabled [3].

- *Two-gap word.* Consecutive envelope-minima gaps take exactly the values $\{4, 5\}$; the 5's occur at minima indices $4, 13, 22, \dots, 139$ — sixteen of them, spaced *exactly* 9, with perfectly period-9 second differences. This is the Sturmian/three-gap signature [20, 21, 16, 17, 18, 19], the block $8 \times 4 + 1 \times 5 = 37$ recurring with envelope period 37.
- *Spectrum.* The FFT peak is at period $4.11 = 1/\alpha$ with power $22.4 \times$ the Bonferroni-corrected 95% red-noise threshold (all other peaks sub-threshold); the ACF fundamental is 4.0 (integer rounding of 4.11), with integer multiples 8, 12, 21, 29, 58, 86.
- *Positivity.* $\mu_n^{(1)} \in [0.066, 51.68]$ over $1 \leq n \leq 600$: deep dips toward zero at sign changes (Obs. 4.2) but $\mu_n^{(1)} > 0$ throughout, matching the exact non-vanishing $h_n^{(1)} \neq 0$ for $n \leq 600$ [5].

8. THE LIMIT OF DURABILITY, REFINED

Fast period $1/\alpha \approx 4.11$: durable for all large n (Thm. 5.1), its driver exactly certifiable (Rmk. 5.2), and now confirmed spectrally to $n = 600$.

Envelope, period 37: *confirmed clean to $n = 600$.* The first second-order defect — a block of seven 4's, i.e. a spacing of 8 (not 9) between consecutive 5's — was anticipated near the next convergent denominator $q_4 = 440$, from the preliminary tail of (2). It has *not* occurred by 600. The reading is not a return to rationality (§6 fixes $\alpha \neq 9/37$) but a sharper one: $9/37$ approximates α better than the tail $[\dots, 1, 11, \dots]$ implied, so the partial quotient following the $9/37$ convergent is effectively large and the first defect is *relocated beyond* 600. Concretely, the empirical frequency of 5's, $16/145 = 0.1103$, sits closer to $1/9 = 0.1111$ than to $\{1/\alpha\} = 0.1122$, the deviation that would have forced an 8-spacing by the twelfth 5 ($n \approx 440$). To pin the relocated defect, refine θ by several more digits and recompute the partial quotient after $9/37$; the next convergent denominator is the predicted scale, testable by extending the envelope run [6].

Strict positivity ($\mu_n^{(c)} > 0$): unconditional for $1 \leq n \leq 600$ — now both by exact arithmetic [5] and by the run's $\mu_n^{(1)} \geq 0.066$; for all large n conditional on (S) and:

Hypothesis 2 (Diophantine non-degeneracy, D). With $\beta = \varphi_c/\pi - \frac{1}{2}$ and $\|\cdot\|$ the distance to the nearest integer, $\|n\alpha + \beta\|$ does not decay geometrically along any subsequence (it suffices that $\|n\alpha + \beta\| \geq c_0 n^{-s}$).

Since $|\cos(n\theta + \varphi_c)| \geq 2\|n\alpha + \beta\|$ and the error in Lemma 3.1 is geometric, (D) forces $h_n^{(c)} \neq 0$ for all large n [5]. The continued fraction (2), with no sign of unbounded partial quotients, makes α a non-Liouville (indeed apparently bounded-type) number, so (D) is expected; and Weyl equidistribution [14, 15] makes $\|n\alpha + \beta\| \rightarrow 0$ infinitely often, so $\mu_n^{(c)}$ is *not* bounded away from 0 — strict positivity, not bounded-away, is the right target [22, 23].

9. STATUS LEDGER

Claim	Status	Source
H analytic on $ t < 1$	rigorous	[7, 9]
sing. of G = zeros of H ; branch point at simple t_0	rigorous	[9, 12]
$t_0 = \rho e^{i\theta}$ dominant, simple, isolated	empirical	Obs. 2.1
$g_m, h_n^{(c)} = C\rho^{-n} \cos(n\theta + \varphi) + \dots$	conditional (S)	[11]
$\mu_n^{(c)} = 0 \iff h_n^{(c)} = 0$	rigorous	Prop. 4.1
$\mu_n^{(c)}$ tracks $ h_n^{(c)} $ (dip to 0.066)	empirical	Obs. 4.2, [3]
oscillation asymptotic/durable, error $\rightarrow 0$	conditional (S)	Thm. 5.1
period/sign pattern certifiable exactly	rigorous	Rmk. 5.2
$\theta/\pi = 0.243183594574493\dots = [0; 4, 8, 1, 11, 2, \dots], \neq 9/37$	settled numerically	§6
fast period $1/\alpha \approx 4.11$ (FFT $22\times$ threshold)	cond. (S)+confirmed	§7
two-gap $\{4, 5\}$ from one frequency	rigorous (three-gap)	[16, 19]
envelope 37 clean to $n = 600$; defect relocated > 600	empirical	§7, 8
$\mu_n^{(1)} > 0, n \leq 600$	rigorous (exact)+run	[5, 3]
$\mu_n^{(c)} > 0$, all large n	conditional (S)+(D)	[5]
$\mu_n^{(c)}$ not bounded away from 0	rigorous (Weyl)	[14]

One-line conclusion. *Conditional on the transfer hypothesis (S) and the observed dominant complex zero $t_0 = \rho e^{i\theta}$, the deformed coefficients grow like ρ^{-n} with phase $e^{in\theta}$, so $\mu_n^{(c)}$ oscillates with a fast period $1/\alpha \approx 4.11$ that is the asymptotic law (it sharpens with n , confirmed spectrally to $n = 600$ at $22\times$ the noise floor) and an envelope of period 37 from the third convergent $9/37$ of the now-settled irrational $\alpha = \theta/\pi = 0.243183594574493\dots$; that envelope runs clean to $n = 600$ with the first second-order defect relocated beyond the current range, the period structure being anchored in the exact rationals $h_n^{(c)}$ and so immune to the precision wall that faked the undeformed plateau.*

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